METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

DATASHEET E-201

E-201 · Feed/effluent interchanger (hot)

DOCUMENT NO.	:	MEOH-V1-ME-DS-003
DOCUMENT TITLE	:	Datasheet E-201
PROJECT	:	Methanol Converter Ver 1 · OTS
STATUS	:	Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

E-201 · Feed/effluent interchanger (hot)

Equipment list MEOH-V1-ME-EQL-001. Values from the simulator model at its design point unless marked assumed. P&ID: MEOH-PID-001.

General		
Service	Feed/effluent interchanger (hot)	MEOH-PID-001
Type	Vertical, single tube pass, true counter-current (floating head or bellows for the tube/shell expansion), HP closures	TEMA (assumed)
Shells	1 (single)	

Performance (model, design point)		
Duty	44.2 MW	P.q201
Hot side in → out	267.6 → 206.4 °C	
Cold side in → out	96.8 → 253.7 °C	
LMTD / F	46.4 K / 1	single tube pass, true counter-current: F = 1 (the cold outlet is above the hot outlet — a temperature cross a 1-2 shell cannot take)
Model UA	952 kW/K	q201 / LMTD (the model uses K.UA201 with flow corrections)
Service U (with fouling)	450 W/m²K	assumed, typical
Installed area	2330 m²	model UA + 10 % margin (assumed)

Design		
Tube side (effluent)	100 bar g / 335 °C	Zone B
Shell side (bed-1 feed)	100 bar g / 335 °C	Zone B
Code	ASME VIII Div. 1 (assumed); TEMA R for exchangers	
Corrosion allowance	3 mm (CS / Cr-Mo parts), 0 (tubes)	assumed

Construction		
Tubes	3192 × OD 19.05 × 2.11 mm × 12.2 m	triangular pitch 23.8 mm
Shell inside diameter	1500 mm	tube count by TEMA bundle factors
Materials	1¼Cr-½Mo SA-387 Gr 11 Cl 2 shell and channel; tubes SA-213 T11	EQL §3
Thickness (pressure side)	53.4 + 3 → 58 mm (ID 1500)	UG-27(c)(1)

Notes		
1	Both sides are loop gas at almost the same pressure: no tube-rupture relief case (RLB).	
2	TV-211 bypasses part of the cold side (TIC-211).	MEOH-PID-001